

Armin Lotfy

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PROFESSIONAL SUMMARY

Ph.D. in Electrical and Computer Engineering with research experience in reinforcement learning, deep learning, computer vision, optimization, and modeling of physical systems. My work includes learning-based methods for EV energy management and battery control, as well as recent research on image embeddings and machine-learning-based testing. I mainly work with Python, PyTorch, TensorFlow, Ray/RLlib, Optuna, MATLAB/Simulink, Git, and Linux. I am interested in applying this background to research problems involving machine learning, simulation, physics, and scientific computing.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher/ Research Lead

Carleton University/ March Network

2026 – Present | Ottawa, Ontario

- Led a research team and coordinated project planning, technical execution, and delivery.
- Automated data-processing and model-evaluation pipelines using Python.
- Performed data analysis, visualization, and comparative model testing.
- Conducted performance, reliability, and comparative testing of large language models.

Battery Systems and Controls Researcher

Carleton University

09/2020 – 09/2025 | Ottawa, Ontario

- Developed reinforcement learning and optimization-based energy management and battery cell-balancing strategies for electric vehicles.
- Modeled lithium-ion batteries, battery packs, electric vehicles, and hybrid energy storage systems using MATLAB/Simulink, Simscape, and Python.
- Evaluated battery performance, including state of charge, cell imbalance, charging and discharging behavior, power demand, energy consumption, and system efficiency.
- Developed battery state-estimation and forecasting pipelines using filtering techniques, CNNs, and LSTMs.
- Designed reinforcement learning controllers using PPO, SAC, and TD3 for battery energy management and cell balancing.
- Applied MPC, robust optimization, genetic algorithms, and particle swarm optimization to battery control and energy-management problems.
- Analyzed battery simulation data and presented technical findings through reports, technical presentations, peer-reviewed publications, and academic and industry discussions.
- Ph.D. thesis: “**Optimization of Energy Management and Battery Cell Balancing in Battery Electric Vehicles Using Reinforcement Learning Techniques.**”

Teaching Assistant

Carleton University

09/2022 – 04/2024 | Ottawa, Ontario

- Led labs and project coaching, guiding implementations on Raspberry Pi and Xilinx FPGA platforms.
- Mentored 600+ students across Mechatronics & Circuits (ECOR 1043/1044, Python) and Switching Circuits & Digital Design (ELEC 2607, VHDL).

SKILLS

Power & Energy Systems

Energy Storage Systems (ESS), Battery Management Systems (BMS), Energy Management Systems (EMS), Equivalent Circuit Modeling (ECM), Lithium-Ion Battery Fundamentals, Battery Degradation Mechanisms, SOC/SOH Estimation, Cell Balancing, Degradation-Aware Control, Power Flow Optimization

Reinforcement Learning & ML

- KNN, SVM, and ML algorithms
- PPO, SAC, TD3, DDPG, MAPPO
- CNN, RNN, LSTM, POMDPs
- Distributed Systems (RLlib)

Programming & Frameworks

- Python, TensorFlow, PyTorch, RLlib, Gymnasium
- NumPy, Pandas, SciPy, Matplotlib, Optuna, scikit-learn
- MATLAB/Simulink
- LangChain, Hugging Face, Ollama, ChromaDB

Software & DevOps

- Git/GitHub, Docker, SQL, Linux, VS Code
- Experiment tracking, reproducible pipelines
- Battery Data Analysis, Time-Series Analysis, Statistical Analysis, Data Visualization, Performance Evaluation

PROJECTS

Agentic Reinforcement Learning Testing Framework

March Network/ Carleton University

Developing an agentic reinforcement learning framework to systematically evaluate the robustness and retrieval performance of embedding models under diverse image augmentation techniques.

- Designing test framework
- Evaluating dataset and statistical tests

POMDP Controller for BEVs with Battery–Supercapacitor HESS [↗](#)

IRES Lab

- Tech Stack: Python, TensorFlow 2, RLlib, Gymnasium, PPO, CNN-LSTM, EKF
- Formulated an uncertainty-aware energy management problem as a POMDP and trained PPO policies to operate reliably under partial observability.
- Developed equivalent circuit models of lithium-ion batteries and supercapacitors in Python, incorporating battery degradation effects.
- Developed real-time SOC/SOH forecasting and estimation using CNN-LSTM models and an EKF for online inference.
- Implemented the full pipeline in Python (RLlib, TensorFlow 2, Gymnasium) and validated in closed-loop using RMSE/MAE for deployment readiness
- Reduced battery degradation by ~2% while maintaining system performance and operational stability.

CTDE-Based Multi-Agent BMS for SOC/SOH Equalization [↗](#)

IRES Lab

- Tech Stack: PPO, Multi-Agent RL, MATLAB/Simulink
- Applied centralized training with decentralized execution (CTDE) using PPO to optimize current allocation while preserving battery health
- Formulated a shared-reward POMDP with belief-state tracking to handle sensor noise and enable joint optimization of efficiency, safety, and degradation
- Modeled lithium-ion battery cells and battery packs in MATLAB/Simulink to evaluate SOC imbalance, battery performance, and degradation.
- Reduced cell-to-cell SOC mismatch from 7% to 0.51% without degrading overall driving performance
- Improved battery capacity degradation by ~39% compared with a single-agent baseline.

RL-Based Energy Management System for BEVs [↗](#)

IRES Lab

- Tech Stack: MATLAB/Simulink, TD3, Multi-Agent RL, Battery Modeling
- Optimized power distribution and efficiency under variable loads by developing a multi-agent RL-based energy management system in MATLAB/Simulink.

- Trained two TD3 agents—one for power-flow regulation and one for SOC balancing—to ensure stable and deployable control.
- Matched DP-level accuracy, achieving a final SOC within 0.3% of a dynamic-programming baseline without prior system knowledge or a full model.

Enhancing Power Sharing in Microgrids [↗](#)

IRES Lab

- Tech Stack: PPO, MATLAB/Simulink, Control Systems
- Developed a PPO-based microgrid controller to optimally share power among distributed resources under dynamic load and renewable generation fluctuations.
- Modeled the control problem as an MDP in MATLAB/Simulink, defining states, actions, constraints, and reward shaping to ensure efficiency and voltage/frequency stability.
- Reduced Integral of Absolute Error (IAE) by 27% in islanded mode and 36% in grid-connected mode compared to a conventional controller.

EDUCATION

Ph.D. in Electrical Engineering

Carleton University

PUBLICATIONS

Image Augmentation as Test Generation for Deep Learning-Based Image Retrieval Systems [↗](#)

Arxiv

08/27/2026

Proximal Policy Optimization with Predictive and Memory-Aware Battery Management for BEVs Using Hybrid Energy Storage and Battery Health-Oriented Rewards [↗](#)

IEEE Transactions on Vehicular Technology

05/18/2026

Centralized Multi-Agent SOC Control for Battery Health Using Proximal Policy Optimization in EVs [↗](#)

IEEE Transactions on Vehicular Technology

12/01/2024

Enhancing Energy Management Strategy for Battery Electric Vehicles: Incorporating Cell Balancing and Multi-Agent Twin Delayed Deep Deterministic Policy Gradient Architecture [↗](#)

IEEE Transactions on Vehicular Technology

12/01/2023

Hierarchic Multi-Agent Energy Management for Extended Driving Range through Battery Cell Balancing [↗](#)

34th IEEE International Symposium on Industrial Electronics (ISIE 2025)

12/01/2024

Cutting-Edge EMS Technologies for EVs and HEVs: Recent Developments and Future Directions [↗](#)

IEEE Access

12/01/2024

A Proximal Policy Optimization-Based Controller for Enhanced Power Sharing in Microgrids [↗](#)

IEEE Access

12/01/2024